

Hayden Fuller
Modern Communication Systems F25
Lab 2
09/29/25

Theory Problems:

1) OFDM theory

a) useful OFDM symbol duration and cyclic prefix duration

$$T_{\text{ofdm}} / (T_{\text{ofdm}} + T_{\text{CP}}) = 0.9$$

$$T_{\text{CP}} = 1/9 * T_{\text{ofdm}}$$

$$T_{\text{CP}} = \tau_{\text{max}} = 4\mu\text{s}$$

$$T_{\text{ofdm}} = 9 T_{\text{CP}} = 36\mu\text{s}$$

$$T_{\text{sym}} = T_{\text{ofdm}} + T_{\text{CP}} = 36 + 4 = 40\mu\text{s}$$

b) subcarrier spacing and num of subcarriers required

$$\Delta f = 1/T_{\text{ofdm}} = 1/36\mu = 27,777.(7) \text{ Hz} = 27.(7)\text{kHz}$$

$$N = B/\Delta f = 5\text{M} * 36\mu = 180$$

c) practical FFT size

$$\text{smallest } 2^n \geq 180$$

256-point FFT

d) Doppler shift

$$v = 360 * 1000 / 3600 = 100\text{m/s}$$

$$f_D = v/c f_c = 100/3 \times 10^8 * 2 * 10^9 = 666.(6)\text{Hz}$$

$$T_c = 1/(2\pi f_D) = 1/(2\pi 667) = 238.7\mu\text{s}$$

$$T_c/T_{\text{sym}} = 238.7\mu\text{s}/40\mu\text{s} = 5.97 \approx 6 \text{ symbols}$$

e) reference signal (pilot) design, illustrate

Can't meet 90% since CP already accounts for 10%, but we can get pretty close

Place comb pilot every 6th subcarrier in frequency and every 3rd OFDM symbol in time

$$\text{pilot density} = 1/(6*3) = 5.56\%$$

$$\text{net overhead} = \text{CP} + \text{pilots} = 15.56\%$$

$$84.44\% \text{ efficiency}$$

plot:

Time →

Symbol\Sc 0 1 2 3 4 5 6 7 8 9 10 11 (... up to 180)

Sym 0 pilot P D D P D D P D D P D D

Sym 1 data D D D D D D D D D D D D

Sym 2 data D D D D D D D D D D D D

Sym 3 data D D D D D D D D D D D D

Sym 4 data D D D D D D D D D D D D

Sym 5 data D D D D D D D D D D D D

Sym 6 data D D D D D D D D D D D D

Sym 7 pilot P D D P D D P D D P D D

...

Legend: P = pilot RE, D = data RE

2) OFDM Orthogonality

a) what i, j is $[b_i, b_j] = 1$

when $i=j$

$$\frac{1}{T_s} \int_0^{T_s} 1 dt = 1$$

b) which condition if $[b_i, b_j] = 0$

$$[b_i, b_j] = \frac{1}{T_s} \frac{e^{j2\pi DfT_s} - 1}{j2\pi Df}$$

$$[b_i, b_j] = 0 \text{ when } e^{j2\pi DfT_s} - 1 = 0$$

$$e^{j2\pi DfT_s} = 1$$

$$2\pi DfT_s = 2\pi k \quad (k \in \mathbb{Z})$$

$$\Delta f = k/T_s$$

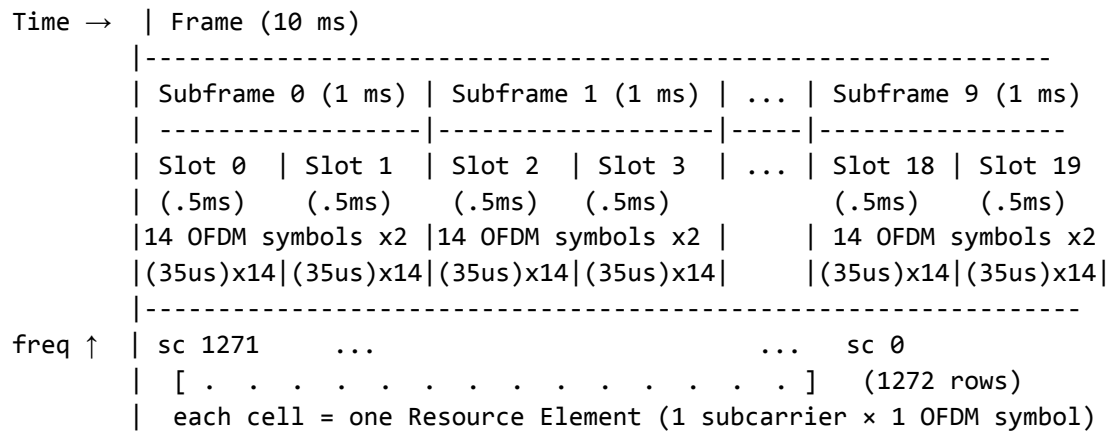
c) minimum subcarrier spacing for orthogonality for given T_s

$$k=1$$

$$\Delta f_{\min} = 1/T_s$$

3) OFDM Example with 5G New Radio

a) 2D time frequency resource grid



b) max theoretical throughput under above assumptions

i) bits per subcarrier

QPSK 2 bits per subcarrier per OFDM symbol

ii) bits per OFDM symbol

1272 subcarriers per OFDM symbol

$1272 * 2 = 2544$ bits per OFDM symbol

iii) total OFDM per second

14 symbols per slot

2 slots per subframe

1000 subframes per second

$2 * 14 * 1000 = 28k$ OFDM symbols per second

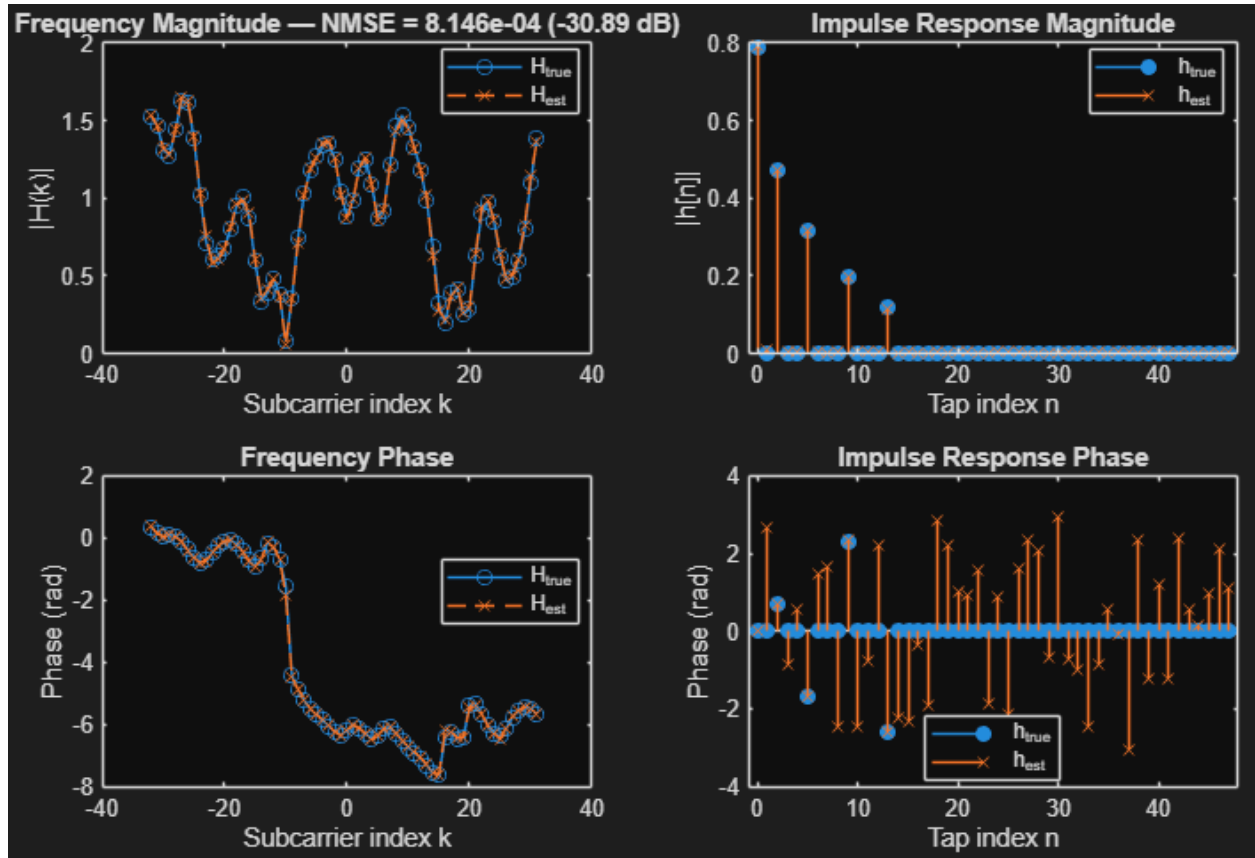
iv) final throughput

$2544 * 28k = 71,232k$ bits/s

71.232Mbps

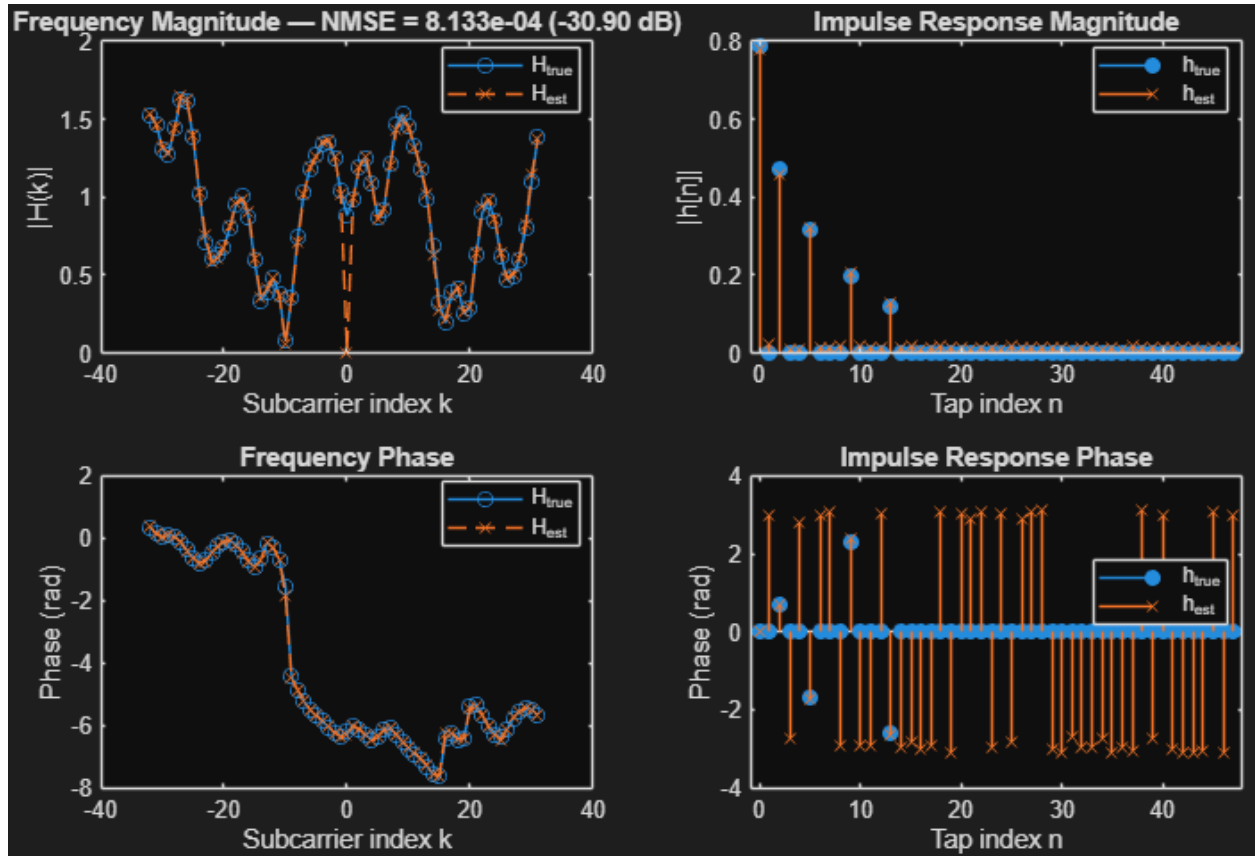
Matlab Simulations:

a) Baseline (clean channel estimate)



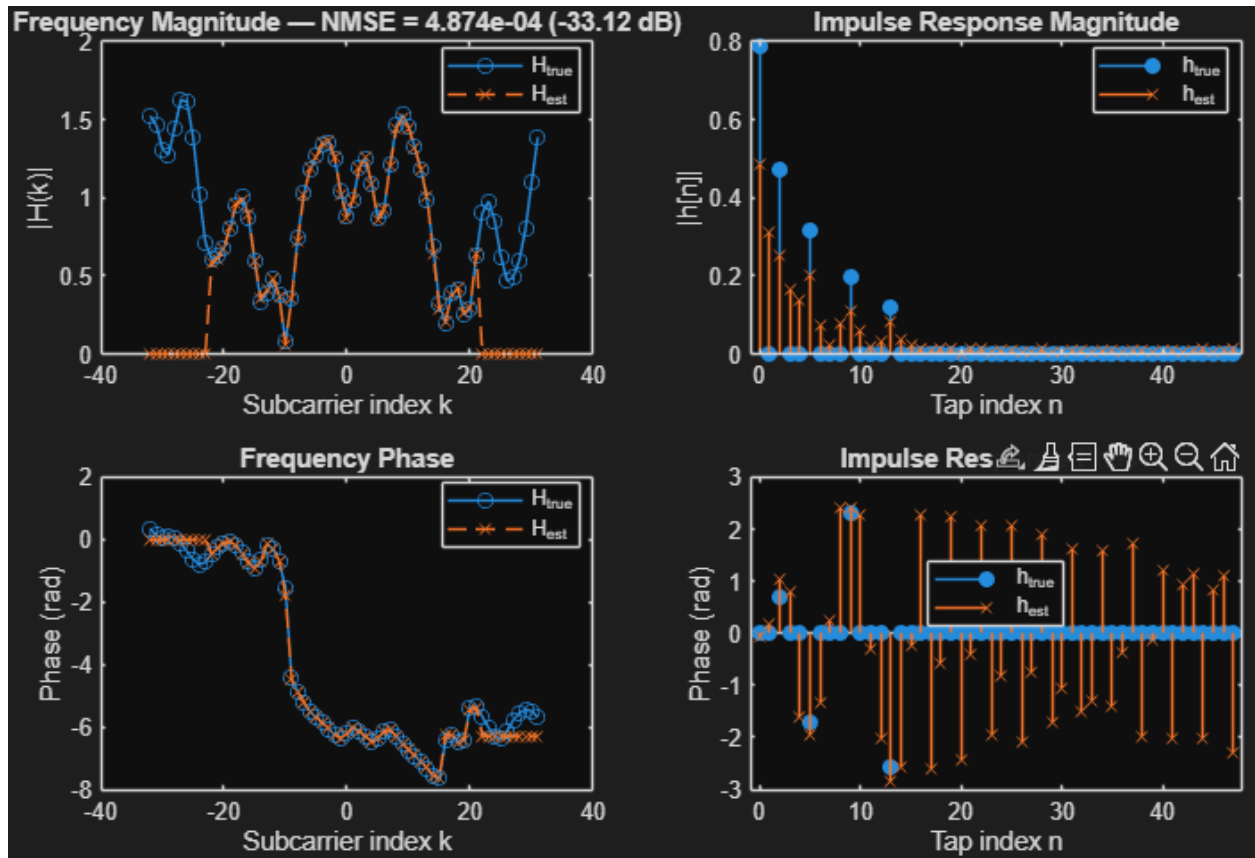
When both $CP \geq L_h - 1$ and the receiver FFT window is perfectly aligned (no timing offset), the system behaves like circular convolution instead of linear convolution, and the per-tone model $Y[k] = H[k]X[k]$ holds exactly for every subcarrier

b) Effect of DC subcarrier nulling



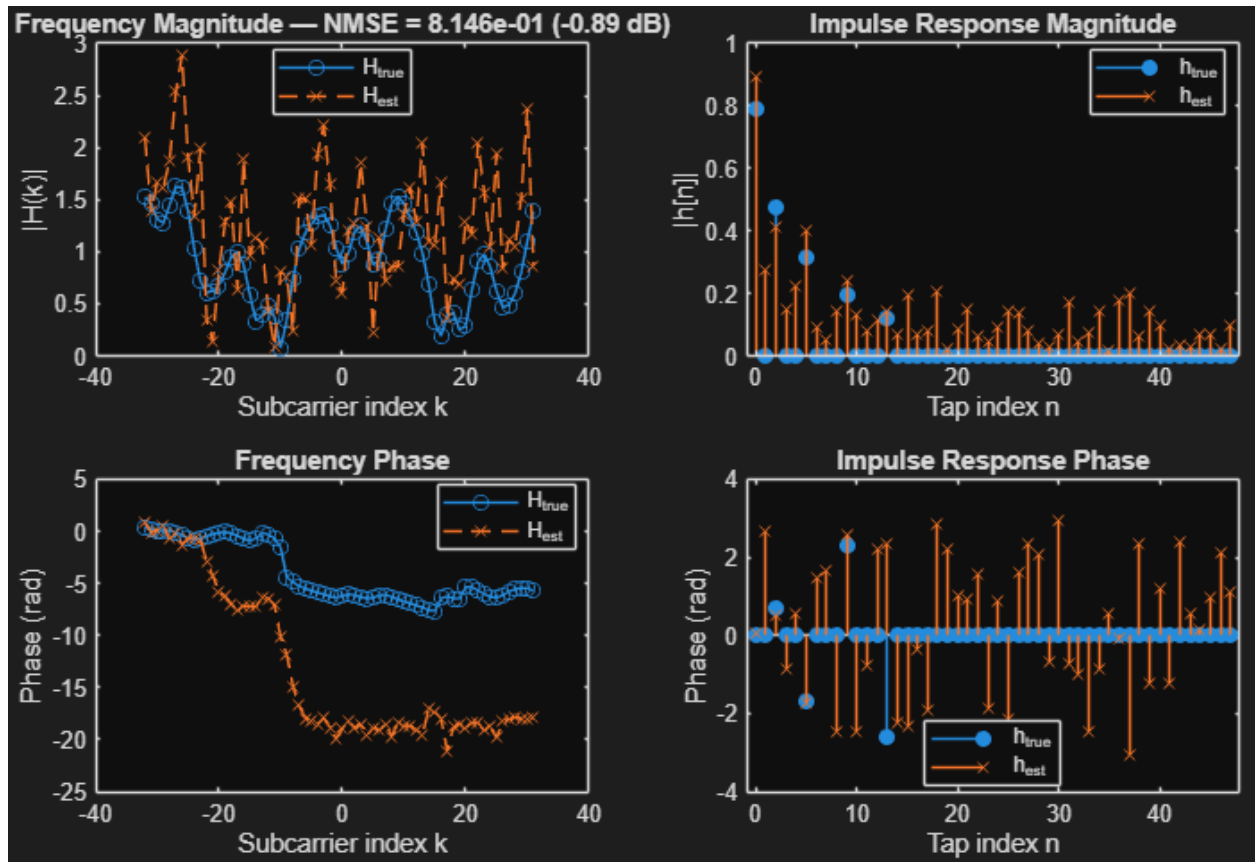
The DC bin is typically nulled in practice because the DC offset in the receiver can distort it badly and the oscillator frequency mismatch can make it unreliable, the simple solution is to null it. This appears on the frequency magnitude plot as a small spike to zero at $k=0$. This only has a small impact on the NMSE, dropping it from 8.146×10^{-4} to 8.133×10^{-4} , a -0.16% change.

c) Guard bands



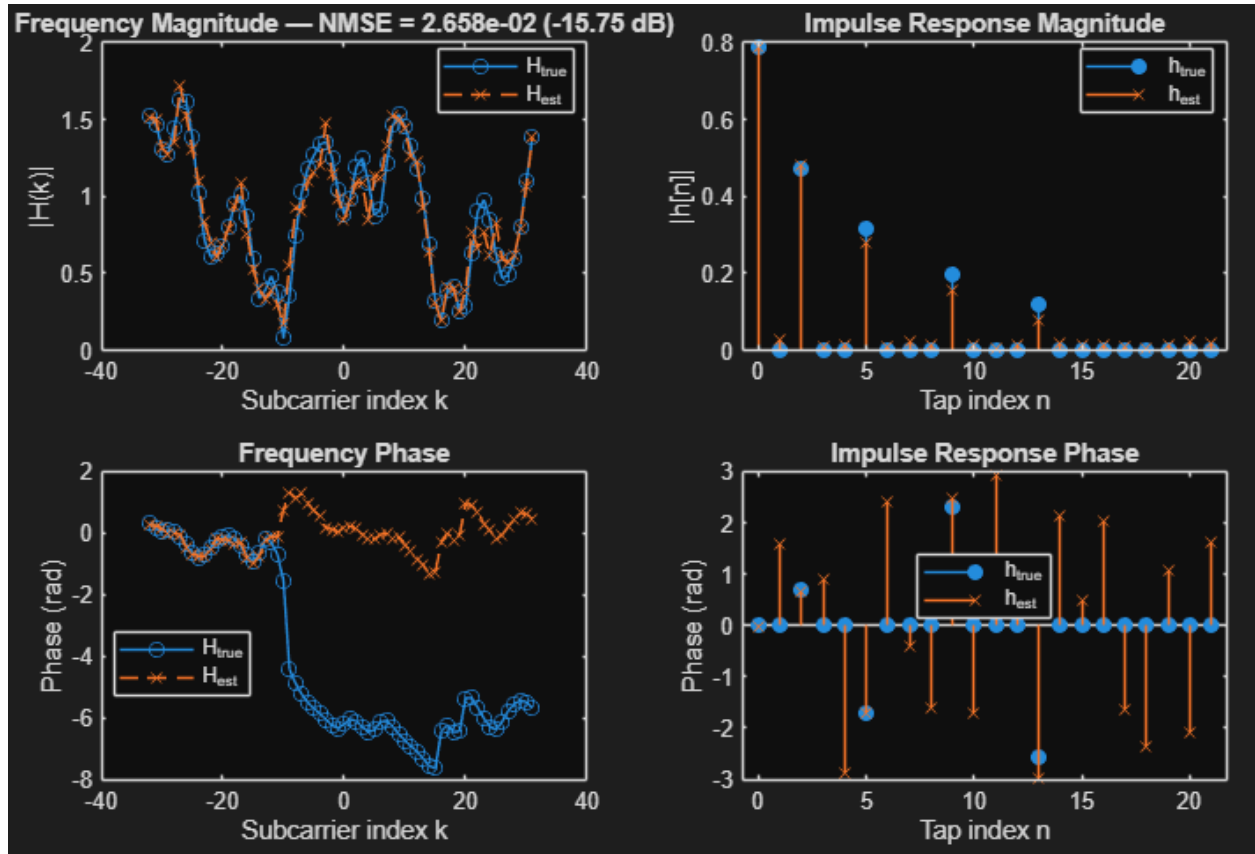
Bins $k \leq -23$ or $k \geq 22$ are nulled, leaving $-23 < k < 22$ as the non-nulled range. Without guards, each subcarrier has a sinc-shaped spectrum, not a perfectly rectangular one, so these sidelobes can leak energy beyond the nominal channel edges and spill into neighboring frequency bands, causing adjacent channel interference. Leaving some subcarriers unused at the edges of the spectrum reduces this leakage and interference. This does have a significant impact on NMSE, dropping it to $4.874e-4$, a -40% change.

d) Noise stress test



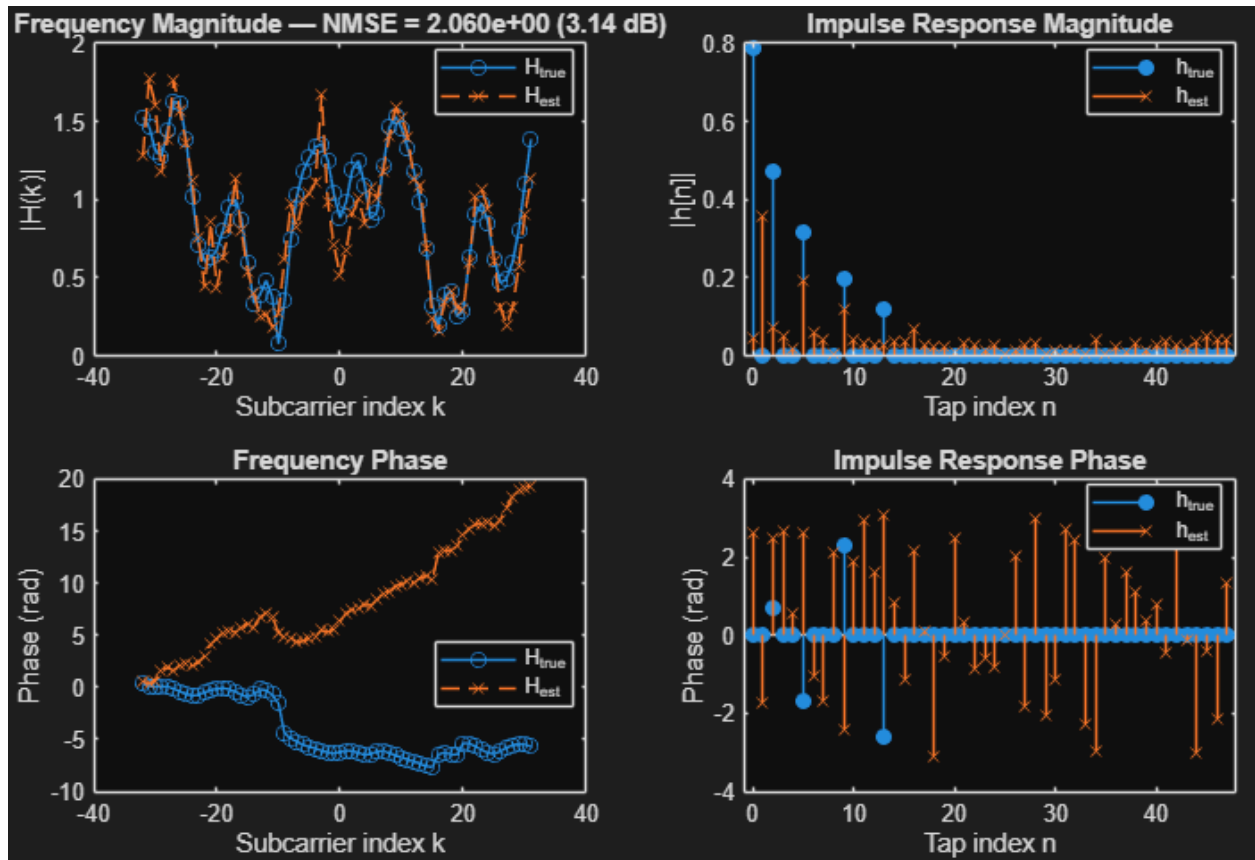
In magnitude, $\hat{H}[k]$ becomes much more jagged with random perturbations compared to $H_{true}[k]$, and in the phase plot, $\hat{H}[k]$ drops off significantly more than $H_{true}[k]$. It enters through the equation $\hat{H}[k] = H_{true}[k] + (N[k]/X[k])$. With this much lower SNR, the NMSE jumps to $8.146e-1$, compared to the previous $8.146e-4$, increasing by 1000x. This results in a largely unreliable estimate.

e) Too-short CP (ISI)



When $CP < L_h - 1$, the system no longer maintains circular convolution. The per-tone model assumes each subcarrier experiences flat fading and remains orthogonal to all others, but if the CP is too short, the signal spills over into the next symbol, causing ISI and ICI. This manifests in the phase plot as $\hat{H}[k]$ becoming distorted and irregular, and in the time plot, the tail of the impulse wrapping around to the beginning, producing interference near $k=0$. The NMSE rises sharply from $8.146e-4$ to $2.658e-2$, about 32.6x.

f) Timing misalignment within CP



The magnitude plot still follows closely, but the phase plot of $\hat{H}[k]$ grows linearly with k . If the receiver starts the window n samples later than the ideal start, the extracted time domain symbol is a circular time shift of the intended symbol. A circular time shift by n samples multiplies each frequency by a linear phase factor. This can be thought of with the fourier dual, a time shift becomes a linear phase in the frequency domain. The NMSE skyrockets to 2.06e0, over 3000x greater than the original.

Below are the changes that were made per run:

Baseline:

```
CP      = 24;  
SNRdB   = 30;  
useDC   = true;  
useGuards = false;  
guard   = 10;  
timing_offset = 0;
```

Effect of DC subcarrier nulling:

```
CP      = 24;  
SNRdB   = 30;  
useDC   = false;  
useGuards = false;  
guard   = 10;  
timing_offset = 0;
```

Guard bands:

```
CP      = 24;  
SNRdB   = 30;  
useDC   = true;  
useGuards = true;  
guard   = 10;  
timing_offset = 0;
```

Noise stress test:

```
CP      = 24;  
SNRdB   = 0;  
useDC   = true;  
useGuards = false;  
guard   = 10;  
timing_offset = 0;
```

Too-short CP (ISI):

```
CP      = 2;  
SNRdB   = 30;  
useDC   = true;  
useGuards = false;  
guard   = 10;  
timing_offset = 0;
```

Timing misalignment within CP:

```
CP      = 24;  
SNRdB   = 30;  
useDC   = true;  
useGuards = false;  
guard   = 10;  
timing_offset = 4;
```

```

% MCS Lab 3 Pilot-only OFDM channel estimation
% Hayden Fuller
% 1) Place pilots on subcarriers (frequency domain)
% 2) IFFT to time, add CP
% 3) Pass through multipath channel (linear convolution) + AWGN
% 4) Remove CP, FFT
% 5) Per-subcarrier channel estimation ( $\hat{H} = Y/X$ )
% 6) Plot true vs. estimated channel in frequency and time domains
% 7) Getv multiple plots for each sub-question

clearvars
rng(123);

%% Parameters
N      = 64;           % FFT size
CP     = 24;           % CP length
SNRdB  = 30;           % SNR (dB); set inf for no noise
useDC  = true;         % set false to remove DC subcarrier
useGuards = false;     % set true to remove Guard bands
guard  = 10;           % guard band #subcarriers one of the two edges

% Multipath channel (discrete-time tapped delay line)
% delays in samples, complex gains
tap_delays = [0 2 5 9 13];
tap_gains  = [1.0 0.6 0.4 0.25 0.15] .* exp(1j*2*pi*[0.00 0.11 -0.27 0.37 -0.41]);
tap_gains  = tap_gains / sqrt(sum(abs(tap_gains).^2)); % normalize power to 1

%% Pilot allocation
% Fill all usable subcarriers with BPSK pilots

pilotF = randi([0 1], N, 1) * 2 - 1; % values in {+1, -1} pilot value on each
subcarrier
if ~useDC
    pilotF(N/2+1) = 0; % DC bin (center) set to zero
end
if useGuards
    pilotF(1:guard) = 0; % here we simply null a few bins at edges
    pilotF(end-guard+1:end) = 0;
end

%% TX: IFFT, CP append
% NOTE: We'll treat 'pilotF' as an FFT-shifted vector for visualization.
% For actual IFFT, we need unshifted frequency ordering.
pilotF_unshift = ifftshift(pilotF); % put DC at index 1 for ifft()
x_noCP = ifft(pilotF_unshift, N); % time-domain symbol (length N)

```

```

x_tx    = [x_noCP(end-CP+1:end); x_noCP];    % prepend CP (length N+CP)

%% Channel
% Build discrete LTI channel impulse response h (length Lh = max delay +1)
Lh = max(tap_delays) + 1;
h = zeros(Lh,1);
h(tap_delays + 1) = tap_gains;

% The "true" frequency response for circular convolution of length N
% is the N-point DFT of h zero-padded to length N, viewed with fftshift.
H_true_unshift = fft([h; zeros(N - numel(h), 1)], N);
H_true = fftshift(H_true_unshift);

freq_axis = -N/2:N/2-1; % shifted freq axis

% For comparison, we plot first, say, 2*CP taps
plot_taps = min(N, max(2*CP, Lh + 8));      % a reasonable window

%% Plot ground truth True channel
figure(1);clf;
subplot(2,2,1);
plot(freq_axis, abs(H_true), 'o-', 'DisplayName','True'); hold on;
xlabel('Subcarrier index k'); ylabel('Magnitude');
title('Frequency Response Magnitude');
legend('Location','northeast');

subplot(2,2,3);
plot(freq_axis, angle(H_true), 'o-', 'DisplayName','True'); hold on;
xlabel('Subcarrier index k'); ylabel('Phase (rad)');
title('Frequency Response Phase');
legend('Location','northeast');

subplot(2,2,2);
stem(0:plot_taps-1, abs([h; zeros(plot_taps-numel(h),1)]), 'filled',
'DisplayName','True'); hold on;
xlabel('Tap index n'); ylabel('Magnitude');
title('Channel Impulse Response');
legend('Location','northeast');

subplot(2,2,4);
stem(0:plot_taps-1, angle([h; zeros(plot_taps-numel(h),1)]), 'filled',
'DisplayName','True'); hold on;
xlabel('Tap index n'); ylabel('Phase Rad');
title('Channel Impulse Response');
legend('Location','northeast');

% All blocks below are TODO

```

```

%% Apply channel TODO
y_conv = conv(x_tx, h);
timing_offset = 0;

%% Add AWGN TODO
if isinf(SNRdB)
    y_rx_full = y_conv;
else
    y_rx_full = awgn(y_conv, SNRdB, 'measured');
end

%% RX: Remove CP, take FFT TODO
% For a single symbol, emulate a perfectly timed receiver that drops CP
% We take the window aligned to the symbol after channel: start after CP
% NOTE: Due to linear convolution, the last (Lh-1) samples are transient.
% We'll take the N samples starting from CP; i.e.; drop the CP and
% transient samples
rx_start = CP + 1 + timing_offset;
y_rx = y_rx_full(rx_start : rx_start + N - 1).';
Y_unshift = fft(y_rx, N);

%% OFDM Channel estimate (estimate in freq domain then take ifft for time domain)
TODO
X_unshift = pilotF_unshift(:);
Y_unshift = Y_unshift(:);

if length(X_unshift) ~= length(Y_unshift)
    error('Length mismatch: X_unshift (%d) vs Y_unshift (%d)', ...
        length(X_unshift), length(Y_unshift));
end

Hhat_unshift = zeros(N,1);
used_idx = abs(X_unshift) > 0;
Hhat_unshift(used_idx,1) = Y_unshift(used_idx,1) ./ X_unshift(used_idx,1);
Hhat_unshift(~used_idx,1) = 0;
Hhat = fftshift(Hhat_unshift);
h_hat_circ = ifft(Hhat_unshift, N);

%% Plot Channel frequency response (magnitude and phase); and Channel impulse
response (magnitude and phase)
% Compare True channel with estimated channel as two legends in each
% subplot. Compute NMSE and display it in the plot. TODO
U = find(used_idx);

```

```

num = sum(abs(Hhat_unshift(U) - H_true_unshift(U)).^2);
den = sum(abs(H_true_unshift(U)).^2);
NMSE_linear = num / den;
NMSE_dB = 10*log10(NMSE_linear);

h_plot = [h; zeros(plot_taps - numel(h), 1)];
hhat_plot = [h_hat_circ(1:plot_taps).'; zeros(max(0, plot_taps -
numel(h_hat_circ)),1)]; % ensure column

figure(2); clf;

% Frequency magnitude
subplot(2,2,1);
plot(freq_axis, abs(H_true), 'o-', 'DisplayName','H_{true}'); hold on;
plot(freq_axis, abs(Hhat), 'x--', 'DisplayName','H_{est}'); hold off;
xlabel('Subcarrier index k'); ylabel('|H(k)|');
title(sprintf('Frequency Magnitude - NMSE = %.3e (%.2f dB)', NMSE_linear, NMSE_dB));
legend('Location','best');

% Frequency phase
subplot(2,2,3);
plot(freq_axis, unwrap(angle(H_true)), 'o-', 'DisplayName','H_{true}'); hold on;
plot(freq_axis, unwrap(angle(Hhat)), 'x--', 'DisplayName','H_{est}'); hold off;
xlabel('Subcarrier index k'); ylabel('Phase (rad)');
title('Frequency Phase');
legend('Location','best');

% Time-domain magnitude (impulse responses)
subplot(2,2,2);
stem(0:plot_taps-1, abs(h_plot), 'filled', 'DisplayName','h_{true}'); hold on;
stem(0:plot_taps-1, abs(hhat_plot), 'x', 'DisplayName','h_{est}'); hold off;
xlabel('Tap index n'); ylabel('|h[n]|');
title('Impulse Response Magnitude');
legend('Location','best');

% Time-domain phase (impulse responses)
subplot(2,2,4);
stem(0:plot_taps-1, angle(h_plot), 'filled', 'DisplayName','h_{true}'); hold on;
stem(0:plot_taps-1, angle(hhat_plot), 'x', 'DisplayName','h_{est}'); hold off;
xlabel('Tap index n'); ylabel('Phase (rad)');
title('Impulse Response Phase');
legend('Location','best');

```